

OGC JSON-FG

Geonovum practical exercise

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Program

10.00 - 10.15	Welcome and introductions
10.15 - 10.30	OGC JSON-FG recap
10.30 - 11.00	Results
11:00 - 11.15	Break
11.15 - 11.45	Discussion
11.30 - 12.00	How to continue?

JSON FG recap

What did we learn on the 25th of June:

- Why JSON-FG is being introduced
- What is JSON-FG more than GeoJSON
- We saw examples
- How it is already implemented in QGIS
- How to validate JSON-FG files/endpoints

What is JSON-FG

- Features and Geometries JSON
- It is an encoding for special geometries
- Extension to GeoJSON
- Backwards compatible

- Established on April 30, 2026:
- <https://www.ogc.org/standards/json-fg/>

Why JSON-FG

- GeoJSON is widely used (OGC API's)
- But, it has its limitations:
 - only WGS84
 - no 3D
 - no standard for datetime attributes
 - no curves, arcs nor measures
 - no reference to schemas and feature type

JSON-FG has it!

Mandatory Extensions

- Ability to use Coordinate Reference Systems (**CRS**) other than WGS 84;
- Ability to encode **temporal** characteristics of a feature;
- A statement identifying the **JSON-FG conformance** classes supported by a JSON-FG feature collection, feature, or geometry.

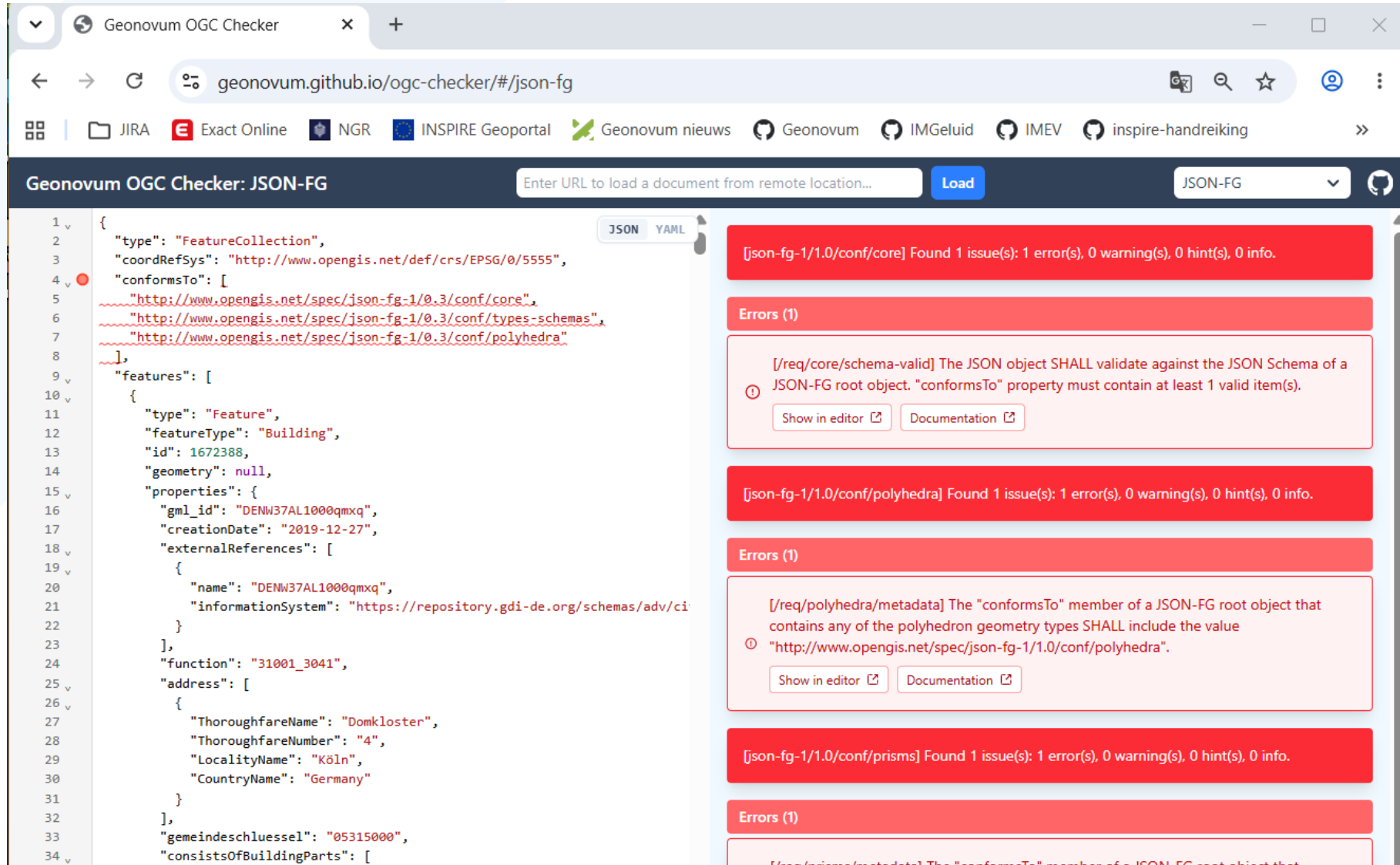
JSON-FG has it!

Optional Extensions with support for:

- **3D:** solids and prisms as geometry types;
- **Complex geometry:**
 - circular arcs
 - compound curves
 - curve polygons;
- Support for **measure** values;
- The ability to declare the **type** and **schema** of a feature and use different feature types in one feature collection.

Validations:

<https://github.com/Geonovum/ogc-checker>



The screenshot displays the Geonovum OGC Checker web application interface. The browser address bar shows the URL `geonovum.github.io/ogc-checker/#/json-fg`. The application header includes a search bar with the text "Enter URL to load a document from remote location..." and a "Load" button. The main content area is divided into two panels. The left panel shows a JSON document being validated, with line numbers 1 through 34 visible. The JSON structure includes a "type" of "FeatureCollection", a "coordRefSys" of "http://www.opengis.net/def/crs/EPSSG/0/5555", and a "features" array containing a single feature with a "type" of "Feature", "featureType" of "Building", "id" of "1672388", and "geometry" of "null". The right panel displays the validation results, showing three error messages. The first error is from the core schema, stating that the "conformsTo" property must contain at least 1 valid item(s). The second error is from the polyhedra schema, stating that the "conformsTo" member of a JSON-FG root object that contains any of the polyhedron geometry types SHALL include the value "http://www.opengis.net/spec/json-fg-1/1.0/conf/polyhedra". The third error is from the prisms schema, stating that the "conformsTo" member of a JSON-FG root object that contains any of the prism geometry types SHALL include the value "http://www.opengis.net/spec/json-fg-1/1.0/conf/prisms".

```
{
  "type": "FeatureCollection",
  "coordRefSys": "http://www.opengis.net/def/crs/EPSSG/0/5555",
  "conformsTo": [
    "http://www.opengis.net/spec/json-fg-1/0.3/conf/core",
    "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas",
    "http://www.opengis.net/spec/json-fg-1/0.3/conf/polyhedra"
  ],
  "features": [
    {
      "type": "Feature",
      "featureType": "Building",
      "id": "1672388",
      "geometry": null,
      "properties": {
        "gml_id": "DENW37AL1000qmxq",
        "creationDate": "2019-12-27",
        "externalReferences": [
          {
            "name": "DENW37AL1000qmxq",
            "informationSystem": "https://repository.gdi-de.org/schemas/adv/ci"
          }
        ],
        "function": "31001_3041",
        "address": [
          {
            "ThoroughfareName": "Domkloster",
            "ThoroughfareNumber": "4",
            "LocalityName": "Köln",
            "CountryName": "Germany"
          }
        ],
        "gemeindeschluessel": "05315000",
        "consistsOfBuildingParts": [

```

[json-fg-1/1.0/conf/core] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/core/schema-valid] The JSON object SHALL validate against the JSON Schema of a JSON-FG root object. "conformsTo" property must contain at least 1 valid item(s).

Show in editor Documentation

[json-fg-1/1.0/conf/polyhedra] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/polyhedra/metadata] The "conformsTo" member of a JSON-FG root object that contains any of the polyhedron geometry types SHALL include the value

"http://www.opengis.net/spec/json-fg-1/1.0/conf/polyhedra".

Show in editor Documentation

[json-fg-1/1.0/conf/prisms] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/prisms/metadata] The "conformsTo" member of a JSON-FG root object that

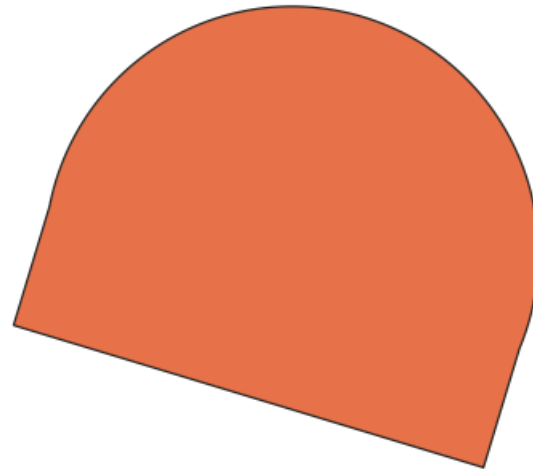
JSON FG practical exercise

What do we want to know:

1. How far are we in NL?
2. What are the obstructions?
3. What is possible and what not?
4. What is needed by software Engineers?
5. Does JSON-FG fulfill all our needs?

What did we ask?

- Make a circular hole in this polygon and keep the old one, while adding a time component



Results

We received 2 results and noticed:

1. Both validated
2. Both could be loaded in QGIS
3. There were some minor differences:
 1. Different order for coordinates of inner ring ([rfc7946 says clock-wise](#))
 2. Identification of the old and new object
 3. Date versus date-time
 4. nr returned: only relevant for API-response, but should be 2
 5. The first line string contained a double point

Results in QGIS

*Praktijkproef_JSON-FG — QGIS

Project Bewerken Beeld Kaartlagen Extra Plug-ins Vector Raster Database Web Mazon Help

Lagen

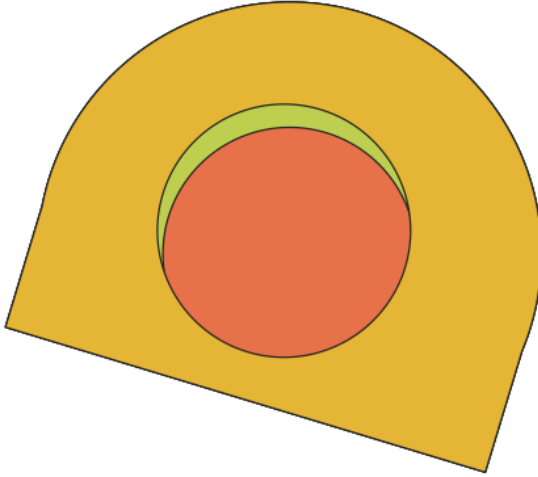
- ☒ **Kadaster BGT FG OndersteunendWegdeel CircularArc — OndersteunendWegdeel**
- ☒ **GISKit_JSON-FG_test_20260922 — OndersteunendWegdeel**
- ☒ **OndersteunendWegdeel_CircularArc — OndersteunendWegdeel**

Tijdbeheer

Huidige frame: 2026-09-22 01:00:00 ≤ t < 2026-09-24 01:00:00

Bereik animatie: 2026-07-10 01:00:00 tot en met 2026-08-23 15:00:00

Tijdbeheer Gereedschappen voor debuggen/ontwikkeling



Logboekmeldingen

Plug-ins X Algemeen X

2026-09-28T08:52:29 INFO Geladen DB Manager (package: db_manager)
2026-09-28T08:52:31 INFO Geladen MetaSearch Catalog Client (package: MetaSearch)

Lagen Browser

Coördinaat 154412,513 463732,638

Schaal 1:24

Vergrootglas 100%

Rotatie 0,0 °

(Her)teken EPSG:28992

Results in Arcgis online

Untitled map

Tables

- OndersteunendWegdeel_CircularArc
- GISkit_JSON_FG_test_20260922
- 07_BGT_FG_OndersteunendWegdeel_CircularArc

Add



Open tabs: 3

- OndersteunendWe...CircularArc
- GISkit_JSON_FG_test_20260922
- 07_BGT_FG_Onder...CircularArc

2 records, 0 selected

	bronhouder	creation_date	functie	functie_codespace	fysiek_voorkomen	fysiek_voorkomen...	in_onderzoek_leeg	kruinlijn_leeg	lokaal_id
	G0307	1/16/2018, 12:00 AM	verkeerseiland	View	open verharding	View	geenWaarde	waardeOnbekend	G0307.92a82176fc03
	G0307	1/16/2018, 12:00 AM	verkeerseiland	View	open verharding	View	geenWaarde	waardeOnbekend	G0307.92a82176fc03

Discussion

What did we want to know:

1. How far are we in NL?
2. What are the obstructions?
3. What is possible and what not?
4. What is needed by software Engineers?
5. Does JSON-FG fulfill all our needs?
6. Is there enough support from the market?

Follow up

- We will publish a small report on the exercise we did
- Presentations and handson remain available on Github.
- Ask your questions via the [GeoForum](#).
- Feedback from participants?
- More workshops to follow, see [Geonovum agenda](#).
- More information:
 - <https://docs.ogc.org/is/21-045r1/21-045r1.html>
 - <https://github.com/opengeospatial/ogc-feat-geo-json>

Thank you!

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